

3D picking for any industrial robot. A depth camera on the wrist and a small computer beside the robot find every part of the size you enter lying on a flat surface, number them in the pick order you choose, and bring the tool to the grip on each one. A connector puts this into the robot's own programs; the robot's program opens and closes the gripper.

Planning figures, not guarantees. The numbers below are deliberately conservative estimates derived from the software's limits, its programmed speeds and the camera's noise. This release is a beta and the figures have not been measured on a production cell. **Test with your parts, gripper and layout before you commit to them.**

COMPATIBLE ROBOTS

Arm	6-axis articulated	Collaborative or industrial. Comes straight down; turns the tool to align the fingers.
Reach	≥ 0.5 m^E	Rated. Every grip is checked against the arm's kinematics first.
Payload	+ 0.1 kg	Camera and bracket, centre of gravity ~40 mm off the flange axis.
Tool flange	ISO 9409-1-50-4-M6 ISO 9409-1-80-6-M8	Brackets for both, under the gripper. Others need an adapter.
Controller	Ethernet, TCP	Its program opens a TCP socket and moves to a computed pose.
Gripper	Two-finger	Any parallel gripper the robot's program opens and closes.

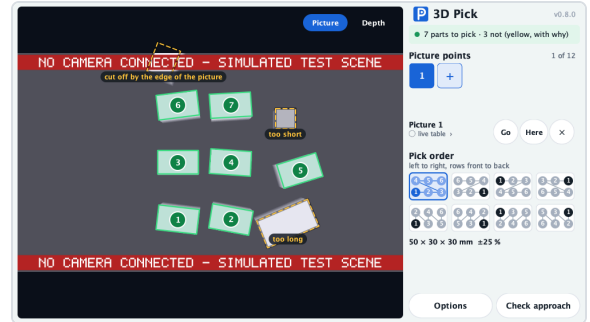
PERFORMANCE (conservative; verify on your cell)

Smallest part	25 × 25 × 15 mm^E	As it lies. The software accepts 5 mm; depth noise makes it unreliable.
Largest part	200 mm long side^E	The whole part must fit in one picture.
Size measurement	±5 mm	Parts that differ by less than 10 mm look alike.
Grip position	±10 mm^E	Plan the jaw opening for it. 4.4 mm RMS calibration in one trial.
Gap between parts	≥ 25 mm^E	20 mm clear where each finger comes down. Touching parts are skipped.
Cycle time, find to grip	15 s per part^E	At the UR connector's moderate speeds. About 2 parts a minute with a short place.
Camera to parts	0.30 to 0.60 m^E	Never closer than 0.25 m.
Area per picture	0.5 × 0.3 m^E	From 0.5 m. Up to 12 pictures per pick.
Surface	Flat, level	Table, tray or bench; parts lying separately.

^E Estimate, not a measured or guaranteed value.

CAPABILITIES

- **Finds parts by volume, from depth alone:** boxes, and cylinders standing on end. Colour and lighting contrast are not used.
- **Any position, any rotation** on the surface. Fingers close across the short side, or the long side of a box if you choose.
- **Eight pick orders:** by rows or by columns, from any corner.
- **Checks before it moves:** reach by the arm's kinematics, and room for the fingers beside the part.
- **Says why not:** rejected parts are drawn in yellow with the reason (off size, touching, out of reach, no finger room, outside the area, cut off).
- **Queues what it saw:** the next part is picked without a trip back to the picture point.
- **Runs inside the robot's own program** through its connector: one step that ends with the tool at the grip. No internet needed.



The Universal Robots connector on a PolyScope 5 pendant (simulated scene).

CONNECTORS

Universal Robots	PolyScope 5.4 to 5.26, URCap 0.9.0 PolyScope X 10.8 to 10.14, URCap 0.7.0
Other makes	On request

WHAT A CELL NEEDS

Camera	RealSense D435 (848 × 480, 30 fps, USB 3) on a printed bracket
Computer	Raspberry Pi class, beside the robot; ships set up
Network	Ethernet; TCP 7621 and 7622
Setup	One camera calibration; optional pick areas
Software	Free, open source (MIT). Kit from Advin.

NOT FOR

- Bin picking: piled or leaning parts
- Shapes other than boxes and upright cylinders
- Transparent or mirror-like parts
- Telling apart parts of the same size
- Driving the gripper or the place move

Download and details: perceptronics.advin.io · **Kit and questions:** nick@advin.io · **Source:** github.com/JimothyJohn/perceptronics

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